Assignment 6

Josh Suber
RIN: 662038070

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Contents

1 Exercise 3.4

(a) We are asked to show that the in-sample estimate of y is given by:

$$\hat{y} = Xw^* + H\epsilon$$

where H is a matrix to be defined.

Solution:

The linear model for generating the target values is given by:

$$y = Xw^* + \epsilon$$

where: - $X \in \mathbb{R}^{N \times d}$ is the design matrix, with N samples and d features, - $w^* \in \mathbb{R}^d$ is the true weight vector, - $\epsilon \in \mathbb{R}^N$ is the noise vector, with $\mathbb{E}[\epsilon] = 0$ and covariance $\mathbb{E}[\epsilon\epsilon^T] = \sigma^2 I$.

The least squares estimate of the weight vector w_{lin} is given by the solution to the normal equations:

$$w_{\text{lin}} = (X^T X)^{-1} X^T y$$

Substitute the expression for $y = Xw^* + \epsilon$ into the equation for w_{lin}:

$$w_{\text{lin}} = (X^T X)^{-1} X^T (Xw^* + \epsilon)$$

Distribute X^T:

$$w_{\text{lin}} = (X^T X)^{-1} (X^T X w^* + X^T \epsilon)$$

Since $(X^T X)^{-1} X^T X = I$, this simplifies to:

$$w_{\text{lin}} = w^* + (X^T X)^{-1} X^T \epsilon$$

Now, the in-sample estimate $\hat{y}$ using w_{lin} is:

$$\hat{y} = X w_{\text{lin}}$$

Substitute $w_{\text{lin}} = w^* + (X^T X)^{-1} X^T \epsilon$ into the expression for $\hat{y}$:

$$\hat{y} = X \left(w^* + (X^T X)^{-1} X^T \epsilon \right)$$

Distribute X:

$$\hat{y} = X w^* + X (X^T X)^{-1} X^T \epsilon$$

Define the matrix $H = X(X^T X)^{-1} X^T$, which is known as the *hat matrix*. Thus, we have:

$$\hat{y} = X w^* + H\epsilon$$

This shows that the in-sample estimate of y is:

$$\hat{y} = X w^* + H\epsilon$$

Final Expression:

$$\boxed{\hat{y} = X w^* + H\epsilon}$$

where $H = X(X^T X)^{-1} X^T$.

(b) We are asked to show that the in-sample error vector $\hat{y} - y$ can be expressed as a matrix times the noise vector ϵ. Specifically, we are looking for the matrix that expresses this relationship.

Solution:

The in-sample estimate of y is given by:

$$\hat{y} = X w_{\text{lin}} = X w^* + H\epsilon$$

where $H = X(X^T X)^{-1} X^T$ is the *hat matrix* and $w_{\text{lin}} = w^* + (X^T X)^{-1} X^T \epsilon$.

The true target vector y is:

$$y = X w^* + \epsilon$$

Now, we want to express the error vector $\hat{y} - y$:

$$\hat{y} - y = (Xw^* + H\epsilon) - (Xw^* + \epsilon)$$

Simplifying:

$$\hat{y} - y = Xw^* + H\epsilon - Xw^* - \epsilon$$

The Xw^* terms cancel out, leaving:

$$\hat{y} - y = H\epsilon - \epsilon$$

Factor out ϵ:

$$\hat{y} - y = (H - I)\epsilon$$

where I is the identity matrix of size $N \times N$.

Thus, the in-sample error vector $\hat{y} - y$ can be expressed as:

$$\hat{y} - y = (H - I)\epsilon$$

Final Expression:

$$\boxed{\hat{y} - y = (H - I)\epsilon}$$

This shows that the in-sample error vector is the product of the matrix $H - I$ and the noise vector ϵ, where H is the hat matrix.

(c) We are asked to express $E_{\text{in}}(w_{\text{lin}})$, the in-sample error, in terms of the noise vector ϵ, and then simplify the expression using the results from *Exercise 3.3(c)*.

Solution:

The in-sample error $E_{\text{in}}(w_{\text{lin}})$ is defined as the average squared error between the predicted values $\hat{y}$ and the true values y:

$$E_{\text{in}}(w_{\text{lin}}) = \frac{1}{N}\|\hat{y} - y\|^2$$

From **Part (b)**, we know that the in-sample error vector $\hat{y} - y$ can be expressed as:

$$\hat{y} - y = (H - I)\epsilon$$

Thus, the squared error becomes:

$$E_{\text{in}}(w_{\text{lin}}) = \frac{1}{N}\|(H - I)\epsilon\|^2$$

The squared norm $\|(H - I)\epsilon\|^2$ is given by:

$$\|(H - I)\epsilon\|^2 = \epsilon^T(H - I)^T(H - I)\epsilon$$

Since H is a symmetric matrix ($H = H^T$), we have:

$$E_{\text{in}}(w_{\text{lin}}) = \frac{1}{N}\epsilon^T(H - I)^2\epsilon$$

We can now use the result from *Exercise 3.3(c)*, which tells us that the expectation of a quadratic form $\epsilon^T A\epsilon$, where ϵ is a noise vector with mean 0 and covariance $\sigma^2 I$, is given by:

$$\mathbb{E}[\epsilon^T A\epsilon] = \sigma^2\text{Tr}(A)$$

In this case, we want the expected value of $E_{\text{in}}(w_{\text{lin}})$. Therefore, we compute the expectation:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \frac{\sigma^2}{N}\text{Tr}((H - I)^2)$$

Now, simplify $(H - I)^2$. Since H is idempotent ($H^2 = H$), we have:

$$(H - I)^2 = H - 2H + I = I - H$$

Thus, the trace becomes:

$$\text{Tr}((H - I)^2) = \text{Tr}(I - H) = \text{Tr}(I) - \text{Tr}(H)$$

The trace of the identity matrix I is N (the number of data points), and the trace of H is the rank of H, which is d (the dimensionality of the data). Therefore:

$$\mathrm{Tr}(I - H) = N - d$$

Finally, we substitute this into the expression for the expected in-sample error:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \frac{\sigma^2}{N}(N - d)$$

Thus, the expected in-sample error simplifies to:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \sigma^2 \left(1 - \frac{d}{N}\right)$$

Final Expression:

$$\boxed{E_D[E_{\text{in}}(w_{\text{lin}})] = \sigma^2 \left(1 - \frac{d}{N}\right)}$$

This shows that the expected in-sample error depends on the dimensionality d of the data, the number of data points N, and the noise variance σ^2.

(d) **Part (d) Solution:**

We are asked to prove that the expected in-sample error $E_D[E_{\text{in}}(w_{\text{lin}})]$ is given by:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \sigma^2 \left(1 - \frac{d+1}{N}\right)$$

This involves showing the influence of the noise terms $\epsilon_1, \ldots, \epsilon_N$ and considering the trace of the matrix in our calculations.

Solution:

From **Part (c)**, we have already derived the following expression for the expected in-sample error:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \frac{\sigma^2}{N}\text{Tr}((H-I)^2)$$

and we simplified $(H-I)^2$ to:

$$(H-I)^2 = I - H$$

which led us to the trace:

$$\text{Tr}(I-H) = N - d$$

Thus, the expected in-sample error at this stage is:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \sigma^2\left(1 - \frac{d}{N}\right)$$

Now, we need to prove that this result should actually be:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \sigma^2\left(1 - \frac{d+1}{N}\right)$$

The extra term $+1$ accounts for the inclusion of the bias term in the linear regression model. In linear regression, the hypothesis function typically includes a bias term w_0, which is the intercept. This bias term behaves like an additional degree of freedom, effectively increasing the number of parameters from d to $d+1$.

To incorporate this extra degree of freedom, we modify the expression for the trace. Specifically, the trace of the matrix H becomes $d+1$ because the rank of H is $d+1$, not just d. Therefore, we adjust the trace as follows:

$$\text{Tr}(I-H) = N - (d+1)$$

Substitute this into the expression for the expected in-sample error:

$$E_D[E_{\text{in}}(w_{\text{lin}})] = \frac{\sigma^2}{N}(N-(d+1)) = \sigma^2\left(1 - \frac{d+1}{N}\right)$$

This shows that the expected in-sample error is:

$$\boxed{E_D[E_{\text{in}}(w_{\text{lin}})] = \sigma^2 \left(1 - \frac{d+1}{N}\right)}$$

Thus, the presence of the bias term increases the dimensionality by 1, leading to the extra +1 in the final expression.

2 Problem 3.1

(a) PLA

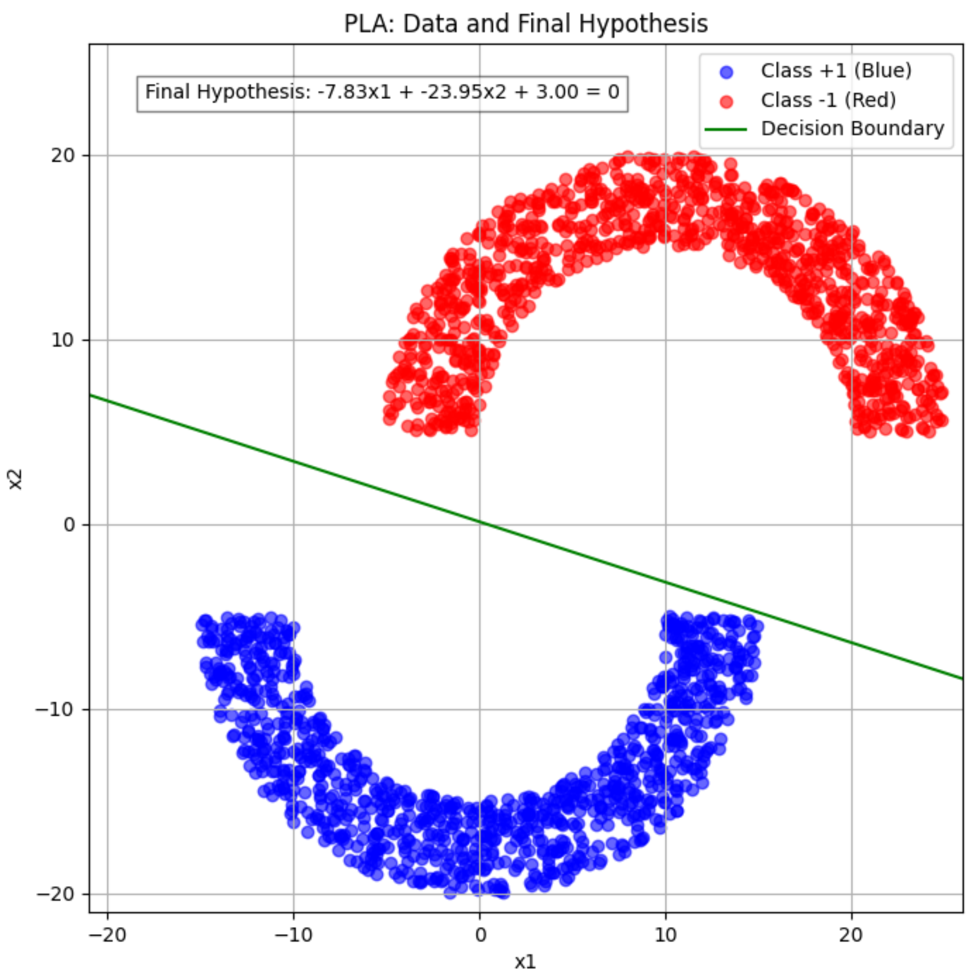

Figure 1: Plot of Data and Final Hypothesis using PLA

(b) Linear Regression

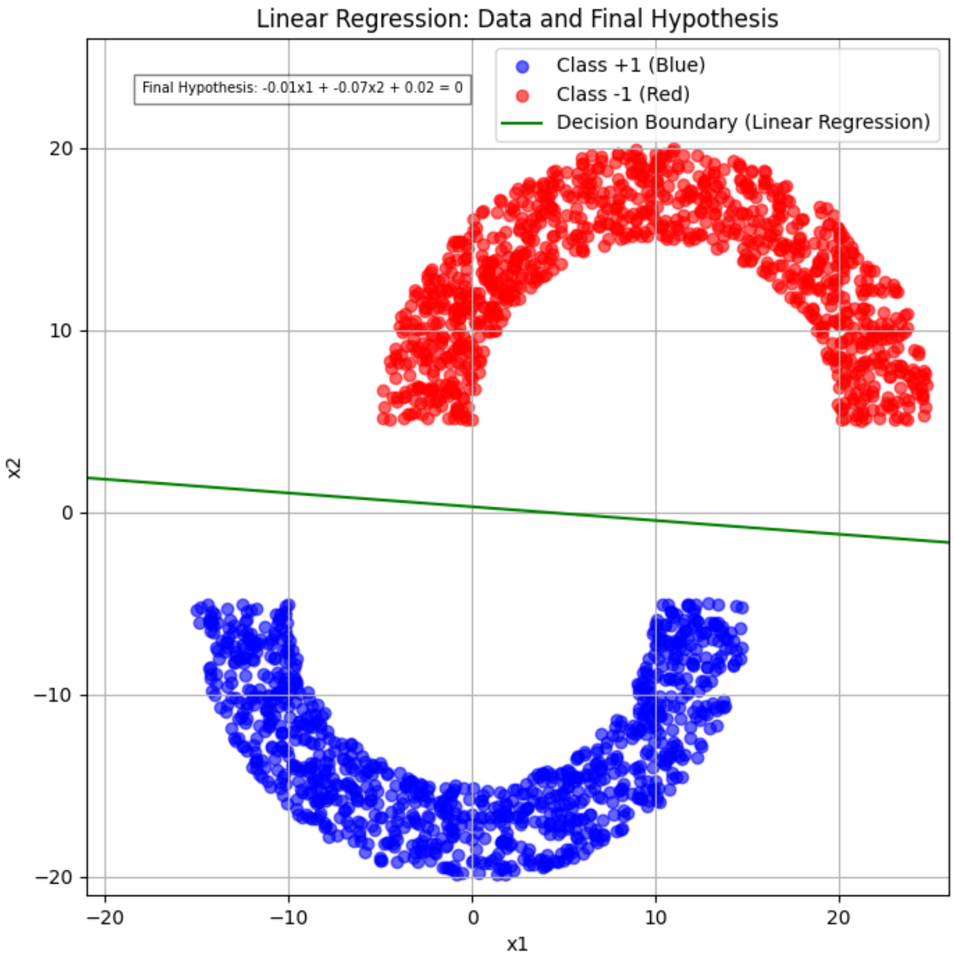

Figure 2: Plot of Data and Final Hypothesis using Linear Regression

3 Problem 3.2

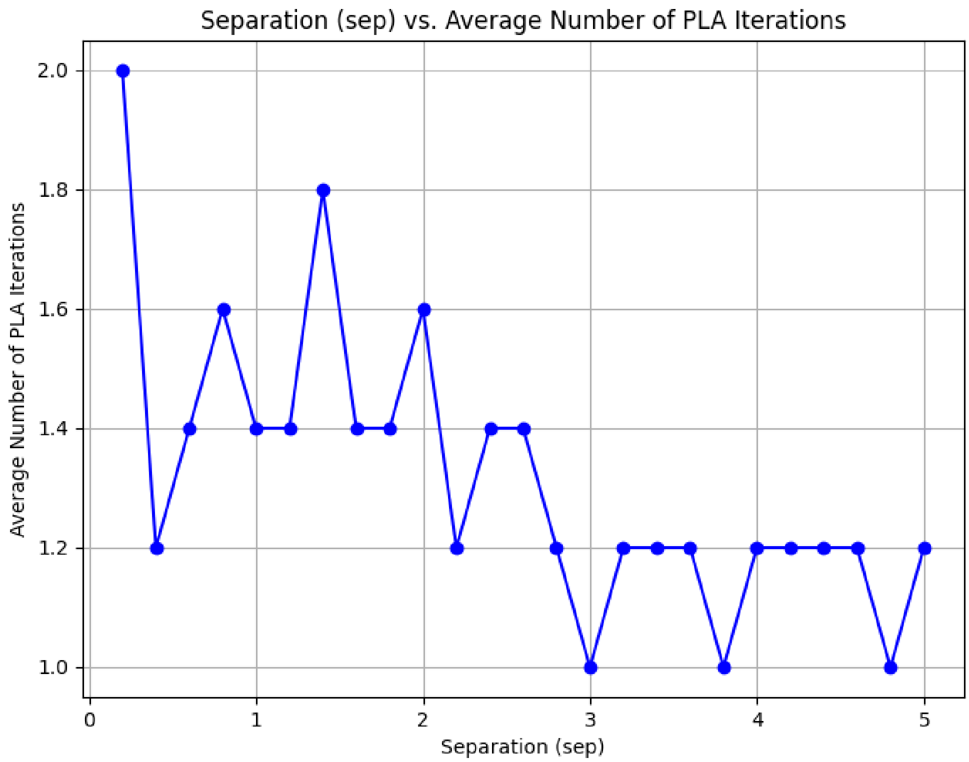

Figure 3: Sep vs Average Number of Iterations for PLA

This plot shows the relationship between the **separation (*sep*)** between the two semi-circles and the **average number of iterations** taken by the Perceptron Learning Algorithm (PLA) to converge. Below are the key observations:

Key Observations:

1. **For small *sep* values (close to 0):**

   PLA takes **more iterations** to converge when the separation is small. When *sep* is close to 0, the two classes (red and blue semi-circles) are very close together or even overlapping, making them harder to separate with a linear boundary. Thus, PLA takes more time to converge as it struggles to find a decision boundary that separates the points.

2. **For larger *sep* values (from *sep* $\approx$ 1 and higher):**

   As *sep* increases, the two classes (red and blue semi-circles) become more linearly separable, and PLA converges faster. The average number of iterations decreases as *sep* increases.

   Around *sep* $= 3$, we see very low values of average iterations (close to 1.0), indicating that PLA converges almost immediately as the classes become well-separated.

3. **Fluctuations in Iterations:**

   While the general trend is that the number of iterations decreases with increasing *sep*, there are some fluctuations in intermediate *sep* values (from *sep* ≈ 0.5 to *sep* ≈ 2.5). These fluctuations are due to randomness in the data generation process. In some trials, PLA may require a few more iterations to adjust the decision boundary.

Conclusion:

- **Small separation** (*sep*) leads to **more iterations** because the data is harder to separate linearly.
- **Large separation** (*sep*) leads to **fewer iterations** because the data is more linearly separable, allowing PLA to converge quickly.

This behavior aligns with the intuition that the difficulty of finding a decision boundary decreases as the classes become more separated in feature space.

4 Problem 3.8

We are given that the out-of-sample error for linear regression is:

$$E_{\text{out}}(h) = \mathbb{E}\left[(h(x) - y)^2\right]$$

and we are asked to show that the hypothesis that minimizes E_{out} is $h^*(x) = \mathbb{E}[y|x]$.

Let's start by expanding the square term in $E_{\text{out}}(h)$:

$$E_{\text{out}}(h) = \mathbb{E}\left[(h(x) - y)^2\right] = \mathbb{E}\left[(h(x) - \mathbb{E}[y|x] + \mathbb{E}[y|x] - y)^2\right]$$

Now, let's rearrange the terms:

$$E_{\text{out}}(h) = \mathbb{E}\left[(h(x) - \mathbb{E}[y|x])^2 + 2(h(x) - \mathbb{E}[y|x])(\mathbb{E}[y|x] - y) + (\mathbb{E}[y|x] - y)^2\right]$$

Since $h(x)$ and $\mathbb{E}[y|x]$ are functions of x, the cross-term $2(h(x) - \mathbb{E}[y|x])(\mathbb{E}[y|x] - y)$ will disappear because:

$$\mathbb{E}\left[(h(x) - \mathbb{E}[y|x])(\mathbb{E}[y|x] - y)\right] = 0$$

This simplifies the expression to:

$$E_{\text{out}}(h) = \mathbb{E}\left[(h(x) - \mathbb{E}[y|x])^2\right] + \mathbb{E}\left[(\mathbb{E}[y|x] - y)^2\right]$$

The second term, $\mathbb{E}\left[(\mathbb{E}[y|x] - y)^2\right]$, is independent of $h(x)$ and represents the irreducible error. The first term, $\mathbb{E}\left[(h(x) - \mathbb{E}[y|x])^2\right]$, depends on $h(x)$.

To minimize $E_{\text{out}}(h)$, we need to minimize this first term, which is minimized when:

$$h(x) = \mathbb{E}[y|x]$$

Therefore, the hypothesis that minimizes E_{out} is:

$$h^*(x) = \mathbb{E}[y|x]$$

Expected Value of $\epsilon(x)$

Next, we are given that $y = h^*(x) + \epsilon(x)$, where $\epsilon(x)$ is an input-dependent noise term, and we are asked to show that $\mathbb{E}[\epsilon(x)] = 0$.

By definition of $\epsilon(x)$, we have:

$$\epsilon(x) = y - h^*(x)$$

Substitute $h^*(x) = \mathbb{E}[y|x]$ into the equation:

$$\epsilon(x) = y - \mathbb{E}[y|x]$$

Now, take the expectation of $\epsilon(x)$ conditioned on x:

$$\mathbb{E}[\epsilon(x)|x] = \mathbb{E}[y - \mathbb{E}[y|x]|x] = \mathbb{E}[y|x] - \mathbb{E}[y|x] = 0$$

Thus, the expected value of $\epsilon(x)$ is:

$$\mathbb{E}[\epsilon(x)] = 0$$

Conclusion

We have shown that:

1. The hypothesis that minimizes the out-of-sample error E_{out} is $h^*(x) = \mathbb{E}[y|x]$.

2. The noise term $\epsilon(x) = y - h^*(x)$ has an expected value of zero, i.e., $\mathbb{E}[\epsilon(x)] = 0$.

5 Handwritten Digits Data - Obtaining Features

(a) plot of two of the digit images

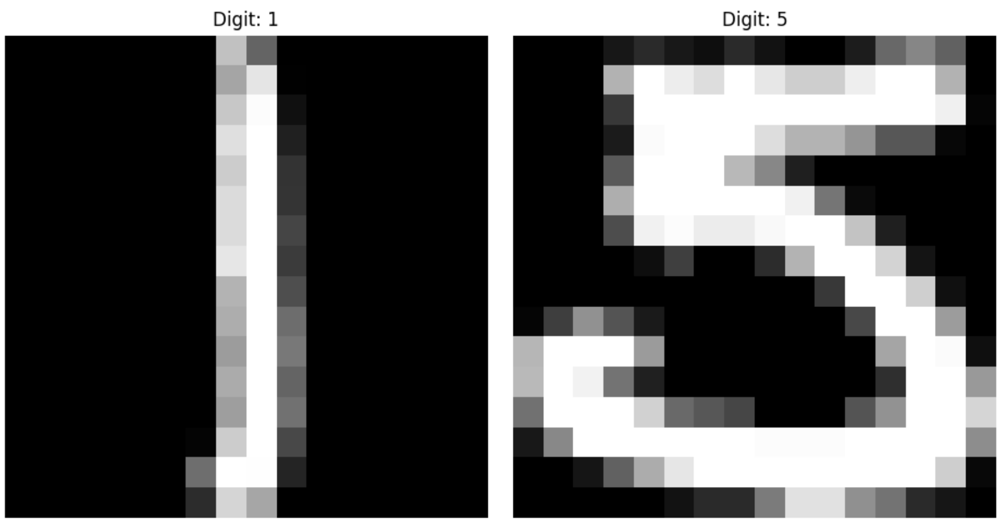

Figure 4: plot of two of the digit images

(b) Features extracted from the digit images

(a) **Average Intensity**

(b) **Vertical Symmetry**

1. Average Intensity

The average intensity measures the overall brightness or darkness of a digit. Since the pixel values in the dataset range from -1 (black) to 1 (white), the average intensity helps determine how much of the image is filled with strokes. Typically, the digit "1" has fewer strokes and, therefore, lower intensity compared to the digit "5", which has more strokes.

Mathematically, the average intensity I is defined as:

$$I = \frac{1}{256} \sum_{i=1}^{256} x_i$$

where x_i is the grayscale value of the i-th pixel, and there are 256 pixels in total (since the image is 16×16).

2. Vertical Symmetry

The vertical symmetry feature measures how symmetric a digit is when reflected about its vertical axis. The digit "1" is typically vertically symmetric, while the digit "5" is asymmetrical due to its loop and tail.

To calculate vertical symmetry, we compare corresponding pixels on the left and right halves of the image. The symmetry score S is computed as:

$$S = 1 - \frac{\sum_{i=1}^{16} \sum_{j=1}^{8} |x_{i,j} - x_{i,16-j+1}|}{\sum_{i=1}^{16} \sum_{j=1}^{16} |x_{i,j}|}$$

where:

- $x_{i,j}$ is the pixel value in row i and column j.
- The numerator computes the absolute differences between corresponding pixels in the left and right halves of the image.
- The denominator normalizes the result by the total pixel intensity in the image.

The vertical symmetry S ranges from 0 (completely asymmetric) to 1 (perfectly symmetric).

Conclusion

These two features, average intensity and vertical symmetry, will help distinguish between the digits 1 and 5. The digit "1" is expected to have a lower average intensity and higher vertical symmetry, while the digit "5" will likely have a higher average intensity and lower vertical symmetry.

```
Digit 1 - Average Intensity: -0.7539140625, Vertical Symmetry: 0.9370023301426584
Digit 5 - Average Intensity: -0.11173828124999999, Vertical Symmetry: 0.6788301608270297
```

Figure 5: Average Intensity and Vertical Symmetry of Digits

(c) Scatter Plot of Features

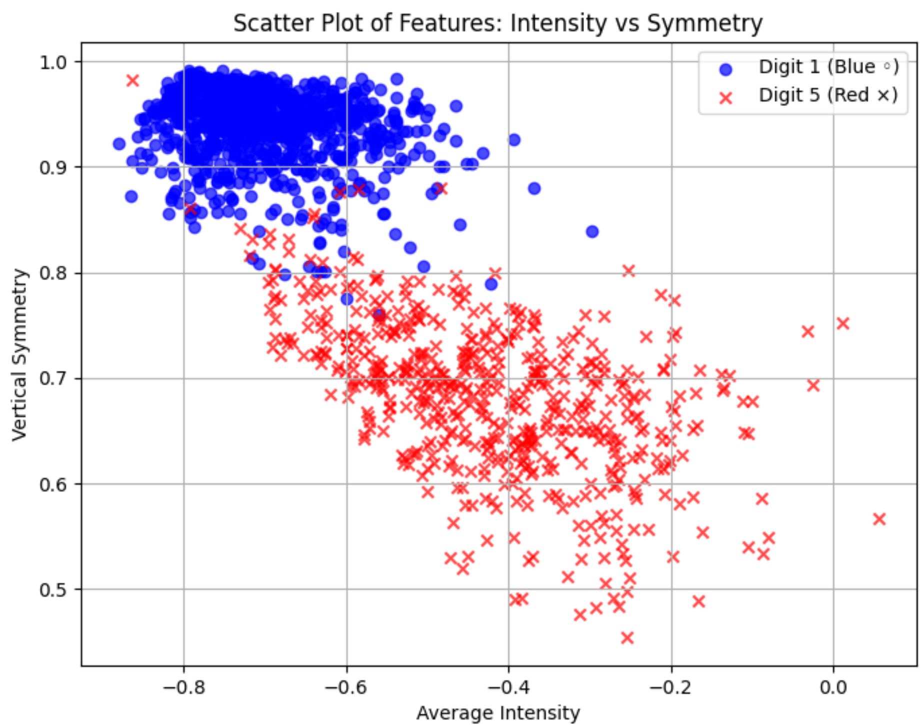

Figure 6: Vertical Symmetry vs Intensity of Digits

The 2-D scatter plot displays the two features, average intensity and vertical symmetry, for the digits 1 and 5. In the plot:

- The blue circles ($\circ$) represent the digit 1.
- The red crosses ($\times$) represent the digit 5.

Observations:

- Digit 1: The blue circles corresponding to digit 1 are clustered in the upper-left region of the plot. This is because digit 1 tends to have:
  - High vertical symmetry: Digit 1 is a single vertical stroke, making it almost perfectly symmetric along the vertical axis.
  - Low average intensity: Digit 1 consists of fewer strokes, so its overall grayscale intensity is lower compared to digit 5.
- Digit 5: The red crosses corresponding to digit 5 are spread out in the lower-right region of the plot. This is because digit 5 typically has:
  - Lower vertical symmetry: The shape of digit 5 includes curves and a tail, which makes it asymmetrical compared to digit 1.
  - Higher average intensity: Digit 5 has more strokes and occupies more space in the image, leading to a higher overall intensity.

Effectiveness of the Features:

The features—average intensity and vertical symmetry—are effective in distinguishing between digits 1 and 5 because the two classes are well-separated in the feature space:

- Vertical symmetry is a strong indicator of whether the digit is 1 (high symmetry) or 5 (low symmetry).
- Average intensity complements the symmetry feature by providing information about how "dark" or "light" the digit is. Since digit 1 is simpler and has fewer strokes, its intensity is lower, while digit 5, being more complex, has a higher intensity.

The clear separation in the scatter plot shows that these two features are useful for classifying the digits 1 and 5 effectively.